

# Yinzhe Zhou

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## EDUCATION

### Johns Hopkins University

*Master of Science in Electrical and Computer Engineering*

Baltimore, MD, USA

*Aug. 2025 – Present*

- **GPA:** 3.73/4.0
- **Relevant Coursework:** FPGA Synthesis Lab, CAD Design of Digital VLSI Systems, Advanced Topics in Microsystem Fabrication, Neural Implants and Interfaces, Mixed-Mode VLSI Systems.

### University of Macau

*Bachelor of Science in Electrical and Computer Engineering*

Macao SAR

*Aug. 2021 – May 2025*

### City University of Hong Kong

*Exchange Student, Department of Electrical Engineering*

Hong Kong SAR

*Jan. 2023 – Aug. 2023*

## PUBLICATIONS

### HACo: Learning Haptic Active Compliance for Force-Aware Dexterous Manipulation

Naisheng Ye, **Yinzhe Zhou**, Junkai Zhao, Yuhang Lu, Checheng Yu, Zhenjie Yang, Pengwei Wang, and Hongyang Li. *arXiv:2609.36596*, 2026.

### OpenTac: An Open Ecosystem for Customizable, Scalable, and Transferable 3D Tactile Sensing

Ariel Slepyan\*, Geeta Chandra Raju Bethala, Jiafeng Gu, Anagha Rajendra Dangle, **Yinzhe Zhou**, Hongyuan Fu, Jade Sim, Tuhin Basu, Xinyi Yin, Md Faizal Karim, Gregory Hager, Homanga Bharadhwaj, Anand Bhattad, Krishna Murthy Jatavallabhula, and Nitish V. Thakor. *Manuscript*, 2026.

## INTERNSHIPS

### Beijing Academy of Artificial Intelligence (BAAI)

*Research Intern*

Beijing, China

*Jun. 2026 – Sep. 2026*

### HACo: Haptic Active Compliance for Dexterous Manipulation

- Developed compliance-regulated teleoperation for dual UR5 arms with Sharpa dexterous hands, combining Cartesian admittance and fingertip force regulation; collected **500 demonstrations** across five tasks with synchronized vision, robot state, tactile, and joint-torque observations.
- Developed a kinematics-aligned haptic encoder fusing ten fingertip tactile streams with 44 hand-joint torques, and gated cross-attention to condition a pretrained vision-language-action model on contact and load feedback.
- Trained the policy with conditional flow matching on compliant-action targets and auxiliary compliant-intent supervision; integrated real-time action chunking for asynchronous closed-loop execution.
- Conducted real-robot comparisons and component ablations across **five force-sensitive tasks**, with 20 trials per task-method pair; HACo achieved **83% mean success**, versus **35%** for the strongest evaluated baseline (**+48 percentage points**).

### Microsoft Research Asia (MSRA)

*Research Intern*

Beijing, China

*Jun. 2024 – Sep. 2024*

### Fine-Tuning Mixtral 8x7B on Biochemical Molecular Data

- Constructed and preprocessed a PubChem-based dataset for molecular property prediction tasks.
- Implemented KNN retrieval over SMILES representations to provide similar-molecule context for in-context learning.
- Fine-tuned Mixtral 8x7B and refined the evaluation protocol, improving test accuracy from **67% to 77%**.
- **Tech:** Python, Git, VS Code.

## RESEARCH EXPERIENCE

### Johns Hopkins University

*Graduate Researcher*

Baltimore, MD, USA

*Feb. 2026 – Present*

### Scalable Shear-Force Sensing for Tactile Perception and Robotic Grasping

- Contributed to a scalable piezoresistive shear-force sensing system spanning multiple sensor form factors and integration onto a UMI handheld gripper for robot data collection.
- Developed embedded firmware and data-acquisition routines for real-time shear-field readout across different sensor-array configurations.
- Designed and ran sensor characterization and robotic grasping experiments to evaluate shear-force features under varied contact conditions.

Johns Hopkins University × TU Darmstadt  
Graduate Researcher

Baltimore, MD, USA  
Nov. 2025 – Present

### Event-Based Visuo-Tactile Sensing for Robot Imitation Learning

- Building a multimodal imitation-learning pipeline that fuses RGB, tactile, and event streams for a UR robot-mounted gripper, targeting contact, slip, and fast-motion cues missed by RGB-only policies.
- Implemented a synchronized multi-rate data pipeline from handheld demonstrations into observation–action samples, aligning RGB at 30 Hz, tactile/event streams at 120 Hz, and 3D pose at 75 Hz.
- Trained and deployed **Diffusion Policy** and **Flow Matching** algorithms on real-world tasks.
- Developing an **event-conditioned Flow Matching** policy with a lightweight 3D-CNN event encoder for high-frequency tactile/event observations.

### University of Macau

Final Year Project — Faculty Dean's Honors List

Macao SAR  
Sep. 2024 – May 2025

### Singular Spectrum Analysis for Biosignal Denoising and Feature Extraction

- Designed a two-level Singular Spectrum Analysis framework combining component correlation and waveform reconstruction for ECG and EEG signal analysis.
- Localized ECG PQRST features and removed EEG artifacts while preserving neural rhythms across delta, theta, alpha, and beta bands.
- Validated the method on synthetic and real recordings, recovering physiological patterns such as eyes-open vs. eyes-closed alpha rhythm differences.
- **Tech:** Python, NumPy, SciPy, Matplotlib, NeuroKit2.

### City University of Hong Kong

Research Assistant

Hong Kong SAR  
May 2023 – Aug. 2023

### Neural Decoding for Implantable Brain–Machine Interfaces with ANN and SNN

- Preprocessed raw primate neural recordings into supervised datasets with movement coordinates, finger positions, and spike trains.
- Designed and benchmarked ANN and SNN decoders for neural decoding, comparing accuracy–efficiency trade-offs for implantable BMI scenarios.
- **Tech:** Python, MATLAB.

### University of Macau

Research Assistant

Macao SAR  
May 2022 – Jul. 2022

### FPGA-Based Hardware Time-Triggered System for Biomedical Engineering

- Implemented FPGA-based signal conditioning and filtering pipelines in Quartus for biomedical data acquisition.
- Refined hardware architecture and I/O interfaces to support deterministic time-triggered behavior.
- **Tech:** Quartus, MATLAB.

## TEACHING EXPERIENCE

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### Johns Hopkins University

Course Assistant, Hands-on Robot Learning (EN.601.498.01.FA26)

Instructor: Krishna Murthy Jatavallabhula.

Baltimore, MD, USA  
Aug. 31 – Dec. 23, 2026

## SELECTED HONORS & AWARDS

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**1st Runner-Up**, IEEE Student Branch Project Competition 2025.

**Akrostar Academic Prize**, University of Macau, 2024/2025.

**Dean's Honor List**, University of Macau, all semesters.

**Academic Excellence Outgoing Exchange Scholarship**, City University of Hong Kong, 2023.

**Top 10%**, Huawei ICT Competition (Preliminary), Macau, 2021–2022.

## TECHNICAL SKILLS

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**Programming:** Python, C/C++, MATLAB, Verilog, VHDL.

**Robotics / ML:** PyTorch, ROS 2, Diffusion Policy, Flow Matching.

**Sensing / Hardware:** Embedded firmware, data acquisition, FPGA development, Vivado, Quartus, Cadence.